Listing 2.7 Using a coordinate transformation matrix to convert local coordinates to projected global coordinates in UTM zone 36S (EPSG:32736).

```
import numpy as np

def create_transformation_matrix(origin_easting,
    origin_northing, rotation_angle_degrees):
  theta = np.radians(rotation_angle_degrees)

  matrix = np.array([ #1
    [np.cos(theta), -np.sin(theta), origin_easting],
    [np.sin(theta), np.cos(theta), origin_northing],
    [0, 0, 1]
  ])

  return matrix

def transform_coordinates(local_coords, transform_matrix):
  homogeneous_coords = np.hstack( #2
    [local_coords,
      np.ones((local_coords.shape[0],
      1))]
  )

  utm_coords = (
    transform_matrix @ homogeneous_coords.T).T #3

  return utm_coords[:, :2] #4

if __name__ == "__main__":
  origin_easting = 471519 #5
  origin_northing = 7977628

  rotation_angle = 30 #6

  transform_matrix = create_transformation_matrix(
    origin_easting, origin_northing, rotation_angle
  )

  local_coords = np.array([ #7
    [0, 0],
    [100, 0],
    [0, 100],
    [100, 100]
```